

```
import pandas as pd

trades = pd.read_csv("historical_data.csv")
sentiment = pd.read_csv("fear_greed_index.csv")
```

```
print(trades.head())
print(sentiment.head())
```

```

      Account  Coin  Execution Price  \
0  0xae5eacaf9c6b9111fd53034a602c192a04e082ed  @107      7.9769
1  0xae5eacaf9c6b9111fd53034a602c192a04e082ed  @107      7.9800
2  0xae5eacaf9c6b9111fd53034a602c192a04e082ed  @107      7.9855
3  0xae5eacaf9c6b9111fd53034a602c192a04e082ed  @107      7.9874
4  0xae5eacaf9c6b9111fd53034a602c192a04e082ed  @107      7.9894

      Size Tokens  Size USD Side  Timestamp IST  Start Position Direction  \
0      986.87    7872.16  BUY    02-12-2024  22:50      0.000000      Buy
1      16.00     127.68  BUY    02-12-2024  22:50    986.524596      Buy
2     144.09    1150.63  BUY    02-12-2024  22:50   1002.518996      Buy
3     142.98    1142.04  BUY    02-12-2024  22:50   1146.558564      Buy
4       8.73     69.75  BUY    02-12-2024  22:50   1289.488521      Buy

      Closed PnL      Transaction Hash  Order ID  \
0      0.0  0xec09451986a1874e3a980418412fcd0201f500c95bac...  52017706630
1      0.0  0xec09451986a1874e3a980418412fcd0201f500c95bac...  52017706630
2      0.0  0xec09451986a1874e3a980418412fcd0201f500c95bac...  52017706630
3      0.0  0xec09451986a1874e3a980418412fcd0201f500c95bac...  52017706630
4      0.0  0xec09451986a1874e3a980418412fcd0201f500c95bac...  52017706630

      Crossed  Fee  Trade ID  Timestamp
0     True  0.345404  8.950000e+14  1.730000e+12
1     True  0.005600  4.430000e+14  1.730000e+12
2     True  0.050431  6.600000e+14  1.730000e+12
3     True  0.050043  1.080000e+15  1.730000e+12
4     True  0.003055  1.050000e+15  1.730000e+12
      timestamp  value  classification  date
0  1517463000     30      Fear  2018-02-01
1  1517549400     15  Extreme Fear  2018-02-02
2  1517635800     40      Fear  2018-02-03
3  1517722200     24  Extreme Fear  2018-02-04
4  1517808600     11  Extreme Fear  2018-02-05
```

```
import pandas as pd
import numpy as np

import matplotlib.pyplot as plt
import seaborn as sns

import warnings
warnings.filterwarnings("ignore")

plt.style.use("ggplot")
```

```
trades = pd.read_csv("historical_data.csv")
sentiment = pd.read_csv("fear_greed_index.csv")
```

```
trades.head()
```

```

      Account  Coin  Execution Price  Size Tokens  Size USD  Side  Timestamp IST  Start Position Direction  Clos
0  0xae5eacaf9c6b9111fd53034a602c192a04e082ed  @107      7.9769    986.87    7872.16  BUY    02-12-2024  22:50      0.000000      Buy
1  0xae5eacaf9c6b9111fd53034a602c192a04e082ed  @107      7.9800     16.00    127.68  BUY    02-12-2024  22:50    986.524596      Buy
2  0xae5eacaf9c6b9111fd53034a602c192a04e082ed  @107      7.9855    144.09    1150.63  BUY    02-12-2024  22:50   1002.518996      Buy
3  0xae5eacaf9c6b9111fd53034a602c192a04e082ed  @107      7.9874    142.98    1142.04  BUY    02-12-2024  22:50   1146.558564      Buy
4  0xae5eacaf9c6b9111fd53034a602c192a04e082ed  @107      7.9894      8.73     69.75  BUY    02-12-2024  22:50   1289.488521      Buy
```

```
trades.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 211224 entries, 0 to 211223
Data columns (total 16 columns):
#   Column                Non-Null Count  Dtype
---  ---
0   Account                211224 non-null object
1   Coin                   211224 non-null object
2   Execution Price        211224 non-null float64
3   Size Tokens            211224 non-null float64
4   Size USD               211224 non-null float64
5   Side                   211224 non-null object
6   Timestamp IST          211224 non-null object
7   Start Position         211224 non-null float64
8   Direction              211224 non-null object
9   Closed PnL             211224 non-null float64
10  Transaction Hash       211224 non-null object
11  Order ID               211224 non-null int64
12  Crossed                211224 non-null bool
13  Fee                    211224 non-null float64
14  Trade ID               211224 non-null float64
15  Timestamp              211224 non-null float64
dtypes: bool(1), float64(8), int64(1), object(6)
memory usage: 24.4+ MB
```

```
trades.describe()
```

	Execution Price	Size Tokens	Size USD	Start Position	Closed PnL	Order ID	Fee	Trade ID	T
count	211224.000000	2.112240e+05	2.112240e+05	2.112240e+05	211224.000000	2.112240e+05	211224.000000	2.112240e+05	2.112240e+05
mean	11414.723350	4.623365e+03	5.639451e+03	-2.994625e+04	48.749001	6.965388e+10	1.163967	5.628549e+14	1.730000e+15
std	29447.654868	1.042729e+05	3.657514e+04	6.738074e+05	919.164828	1.835753e+10	6.758854	3.257565e+14	8.680000e+14
min	0.000005	8.740000e-07	0.000000e+00	-1.433463e+07	-117990.104100	1.732711e+08	-1.175712	0.000000e+00	1.680000e+14
25%	4.854700	2.940000e+00	1.937900e+02	-3.762311e+02	0.000000	5.983853e+10	0.016121	2.810000e+14	1.740000e+14
50%	18.280000	3.200000e+01	5.970450e+02	8.472793e+01	0.000000	7.442939e+10	0.089578	5.620000e+14	1.740000e+14
75%	101.580000	1.879025e+02	2.058960e+03	9.337278e+03	5.792797	8.335543e+10	0.393811	8.460000e+14	1.740000e+14
max	109004.000000	1.582244e+07	3.921431e+06	3.050948e+07	135329.090100	9.014923e+10	837.471593	1.130000e+15	1.750000e+15

```
trades.isnull().sum()
```

```

0
Account      0
Coin         0
Execution Price  0
Size Tokens  0
Size USD     0
Side         0
Timestamp IST 0
Start Position 0
Direction    0
Closed PnL   0
Transaction Hash 0
Order ID     0
Crossed      0
Fee          0
Trade ID     0
Timestamp    0

```

```
dtype: int64
```

```
trades.duplicated().sum()
```

```
np.int64(0)
```

```
# Convert to datetime

trades["Timestamp"] = pd.to_datetime(trades["Timestamp"])

sentiment["date"] = pd.to_datetime(sentiment["date"])
```

```
trades["Date"] = trades["Timestamp"].dt.date

sentiment["Date"] = sentiment["date"].dt.date
```

```
trades[["Timestamp", "Date"]].head()
```

	Timestamp	Date
0	1970-01-01 00:28:50	1970-01-01
1	1970-01-01 00:28:50	1970-01-01
2	1970-01-01 00:28:50	1970-01-01
3	1970-01-01 00:28:50	1970-01-01
4	1970-01-01 00:28:50	1970-01-01

```
sentiment[["date", "Date"]].head()
```

	date	Date
0	2018-02-01	2018-02-01
1	2018-02-02	2018-02-02
2	2018-02-03	2018-02-03
3	2018-02-04	2018-02-04
4	2018-02-05	2018-02-05

```
df = pd.merge(
    trades,
    sentiment[["Date", "classification"]],
    on="Date",
    how="left"
)
```

```
df["Win"] = df["Closed PnL"] > 0
```

```
summary = pd.DataFrame({

    "Metric":[
        "Total Trades",
        "Unique Traders",
        "Average Profit",
        "Median Profit",
        "Maximum Profit",
        "Maximum Loss"
    ],

    "Value":[
        len(df),
        df["Account"].nunique(),
        df["Closed PnL"].mean(),
        df["Closed PnL"].median(),
        df["Closed PnL"].max(),
        df["Closed PnL"].min()
    ]

})

summary
```

	Metric	Value
0	Total Trades	211224.000000
1	Unique Traders	32.000000
2	Average Profit	48.749001
3	Median Profit	0.000000
4	Maximum Profit	135329.090100
5	Maximum Loss	-117990.104100

Next steps: [Generate code with summary](#) [New interactive sheet](#)

```
# Convert Timestamp IST to datetime
trades["Timestamp IST"] = pd.to_datetime(
    trades["Timestamp IST"],
    format="%d-%m-%Y %H:%M"
)

# Convert sentiment date
sentiment["date"] = pd.to_datetime(sentiment["date"])

# Create Date columns
trades["Date"] = trades["Timestamp IST"].dt.date
sentiment["Date"] = sentiment["date"].dt.date
```

```
df = pd.merge(
    trades,
    sentiment[["Date", "classification"]],
    on="Date",
    how="left"
)
```

```
import matplotlib.pyplot as plt

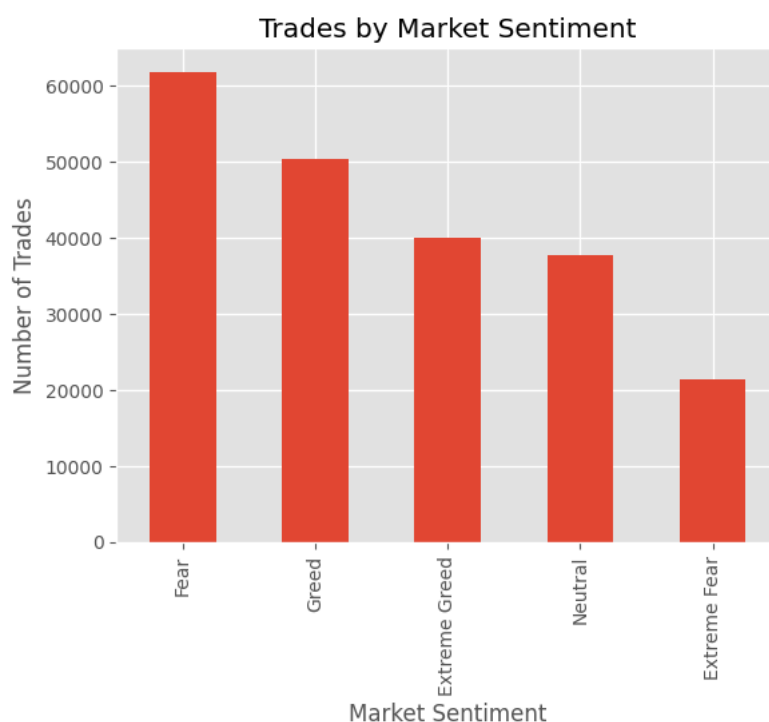
df["classification"].value_counts().plot(kind="bar")

plt.title("Trades by Market Sentiment")

plt.xlabel("Market Sentiment")

plt.ylabel("Number of Trades")
plt.savefig("Trades by market.png", dpi=300, bbox_inches="tight")

plt.show()
```

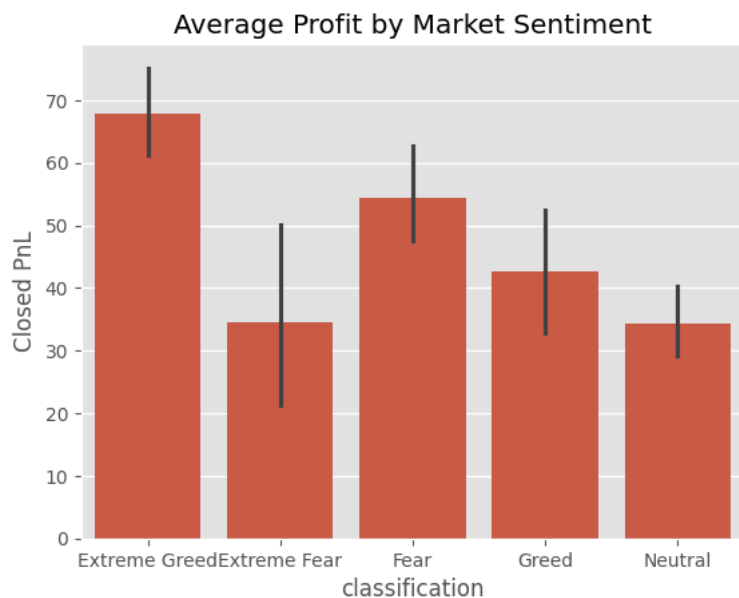


```
import seaborn as sns
```

```
sns.barplot(
    data=df,
    x="classification",
    y="Closed PnL"
)

plt.title("Average Profit by Market Sentiment")
plt.savefig("Average Profit by Market Sentiment.png", dpi=300, bbox_inches="tight")

plt.show()
```



```
coin_profit = (
    df.groupby("Coin")["Closed PnL"]
      .mean()
      .sort_values(ascending=False)
)

coin_profit
```

	Closed PnL
Coin	
@109	270.704207
AVAX	239.095613
ENA	219.524751
@85	200.804091
EIGEN	197.063189
...	...
S	-60.033651
IO	-62.198609
@147	-70.126539
TRUMP	-190.012973
@135	-5981.363990

246 rows × 1 columns

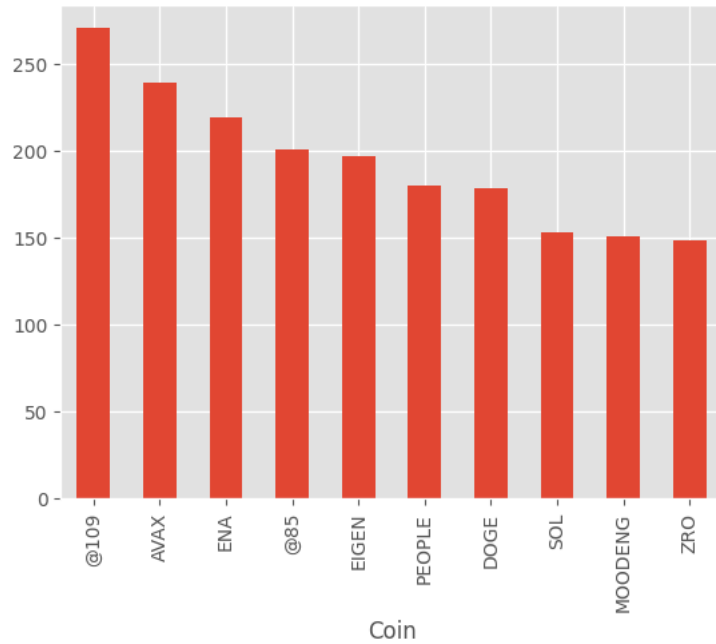
dtype: float64

```
coin_profit.head(10).plot(kind="bar")

plt.title("Top 10 Coins by Average Profit")
plt.savefig("top 10 coin by avg profit.png", dpi=300, bbox_inches="tight")

plt.show()
```

Top 10 Coins by Average Profit



```
# Create a Win column
df["Win"] = df["Closed PnL"] > 0
```

```
df[["Closed PnL", "Win"]].head()
```

	Closed PnL	Win	
0	0.0	False	
1	0.0	False	
2	0.0	False	
3	0.0	False	
4	0.0	False	

```
df.groupby("classification")["Win"].mean() * 100
```

	Win
classification	
Extreme Fear	37.060748
Extreme Greed	46.494299
Fear	42.076750
Greed	38.482794
Neutral	39.699093

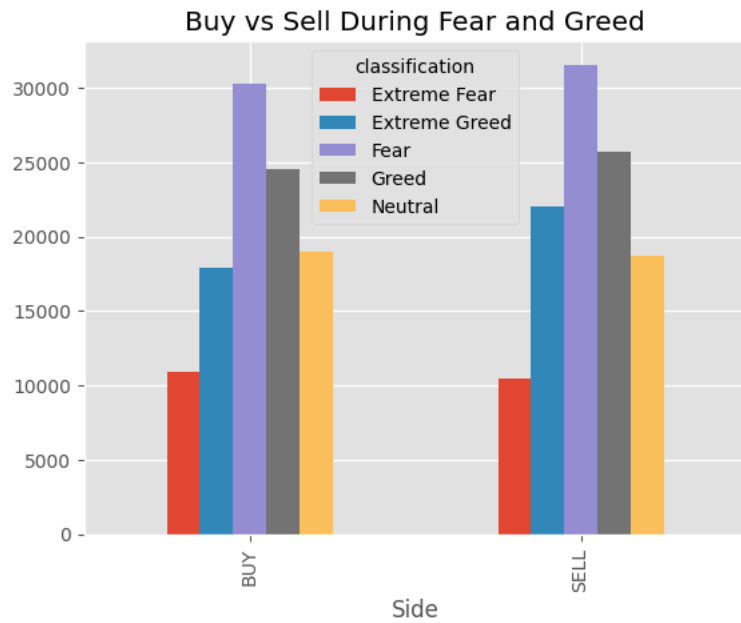
```
dtype: float64
```

```
pd.crosstab(
    df["Side"],
    df["classification"]
)
```

classification	Extreme Fear	Extreme Greed	Fear	Greed	Neutral	
Side						
BUY	10935	17940	30270	24576	18969	
SELL	10465	22052	31567	25727	18717	

```
pd.crosstab(
    df["Side"],
    df["classification"]
).plot(kind="bar")
```

```
plt.title("Buy vs Sell During Fear and Greed")
plt.savefig("buy vs sell", dpi=300, bbox_inches="tight")
plt.show()
```



```
df.groupby("classification")["Size USD"].mean()
```

Size USD	
classification	
Extreme Fear	5349.731843
Extreme Greed	3112.251565
Fear	7816.109931
Greed	5736.884375
Neutral	4782.732661

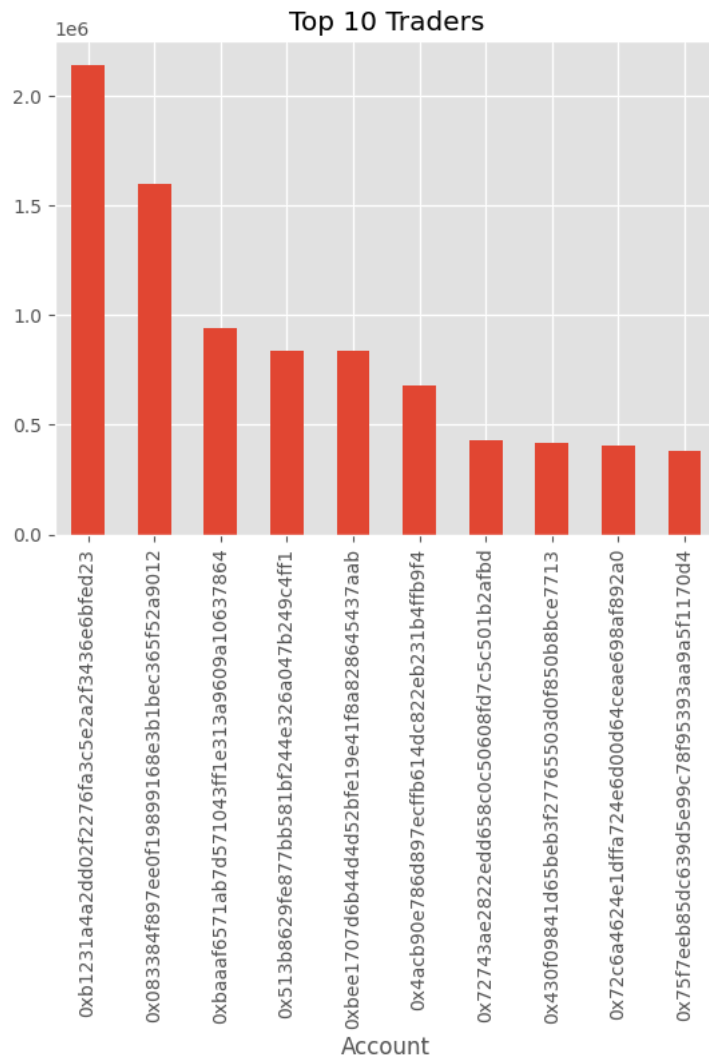
dtype: float64

```
top = (
    df.groupby("Account")["Closed PnL"]
        .sum()
        .sort_values(ascending=False)
        .head(10)
)

top.plot(kind="bar")

plt.title("Top 10 Traders")
plt.savefig("top 10 traders.png", dpi=300, bbox_inches="tight")

plt.show()
```



```
numeric = df.select_dtypes(include="number")
```

```
numeric.corr()
```

	Execution Price	Size Tokens	Size USD	Start Position	Closed PnL	Order ID	Fee	Trade ID	
Execution Price	1.000000	-0.017185	0.189855	0.017241	-0.006505	0.125850	0.225265	0.001841	
Size Tokens	-0.017185	1.000000	0.013310	0.044660	0.005068	-0.043753	0.006710	-0.003848	
Size USD	0.189855	0.013310	1.000000	0.007787	0.123589	0.045210	0.745939	0.000245	
Start Position	0.017241	0.044660	0.007787	1.000000	0.004066	-0.082799	0.010559	0.001165	
Closed PnL	-0.006505	0.005068	0.123589	0.004066	1.000000	-0.007250	0.084020	0.003466	
Order ID	0.125850	-0.043753	0.045210	-0.082799	-0.007250	1.000000	0.039303	0.003880	
Fee	0.225265	0.006710	0.745939	0.010559	0.084020	0.039303	1.000000	0.000940	
Trade ID	0.001841	-0.003848	0.000245	0.001165	0.003466	0.003880	0.000940	1.000000	

```
plt.figure(figsize=(10,8))

sns.heatmap(
    numeric.corr(),
    annot=True,
    cmap="coolwarm"
)
plt.savefig("coolwarm.png", dpi=300, bbox_inches="tight")
plt.show()
```